Table 64. SPI characteristics

Symbol	Parameter[1]	Conditions	Min	Typ	Max	Unit
f_{SCK} $1/t_{c(SCK)}$	SPI clock frequency	Master mode $2\ V < V_{DD} < 3.6\ V$	-	-	24	MHz
		Slave receiver mode			24	
		Slave transmitter mode/full duplex[2] $2.7\ V < V_{DD} < 3.6\ V$			24	
		Slave transmitter mode/full duplex[2] $2\ V < V_{DD} < 3.6\ V$			22	
$t_{su(NSS)}$	NSS setup time	Slave mode	$4 * T_{PCLK}$	-	-	ns
$t_{h(NSS)}$	NSS hold time	Slave mode	$2 * T_{PCLK}$	-	-	ns
$t_{w(SCKH)}$ $t_{w(SCKL)}$	SCK high and low times	Master mode	T_{SCK2}[3] - 1	T_{PCLK}	T_{SCK2}[3] + 1	ns
$t_{su(MI)}$	Data input setup time in master mode	-	4.5	-	-	ns
$t_{su(SI)}$	Data input setup time in slave mode	-	2.5	-	-	ns
$t_{h(MI)}$	Data input hold time in master mode	-	2.5	-	-	ns
$t_{h(SI)}$	Data input hold time in slave mode	-	3	-	-	ns
$t_{a(SO)}$	Data output access time in slave mode	-	10	-	34	ns
$t_{dis(SO)}$	Data output disable time in slave mode	-	9	-	16	ns
$t_{v(SO)}$	Data output valid time in slave mode	$2.7\ V < V_{DD} < 3.6\ V$	-	12	16	ns
		$2\ V < V_{DD} < 3.6\ V$	-	12	22	
$t_{v(MO)}$	Data output valid time in master mode	-	-	3	5.5	ns
$t_{h(SO)}$	Data output hold time in slave mode	-	10	-	-	ns
$t_{h(MO)}$	Data output hold time in master mode	-	1.5	-	-	ns

1. Evaluated by characterization. Not tested in production.
2. Maximum frequency in Slave transmitter mode is determined by the sum of $t_{v(SO)}$ and $t_{su(MI)}$ which has to fit into SCK low or high phase preceding the SCK sampling edge. This value can be achieved when the SPI communicates with a master having $t_{su(MI)} = 0$ while Duty(SCK) = 50%
3. $T_{SCK2} = T_{PCLK} *$ prescaler / 2